# Experiment - 14

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Aim - To study Amazon RDS, create a database instance on AWS RDS, understand security groups, and connect a cloud RDS database to a local database client such as pgAdmin.

Software used –

- AWS Management Console / AWS account
- Amazon RDS
- Amazon EC2 (for the MySQL-client approach shown in the guidelines)
- pgAdmin (for the PostgreSQL local-connection approach)
- Internet connection

Theory - Amazon RDS is a managed relational database service provided by AWS. The guidelines focus on scalability, performance, advanced security, costeffectiveness, manageability, availability and durability.

The experiment contrasts databases running on EC2 with RDS. Running a database on EC2 can provide OS access, advanced DB option tuning, vendor-specific requirements and versions not provided by AWS, but it also introduces administration, backup/DR, monitoring and setup overhead.

Features –

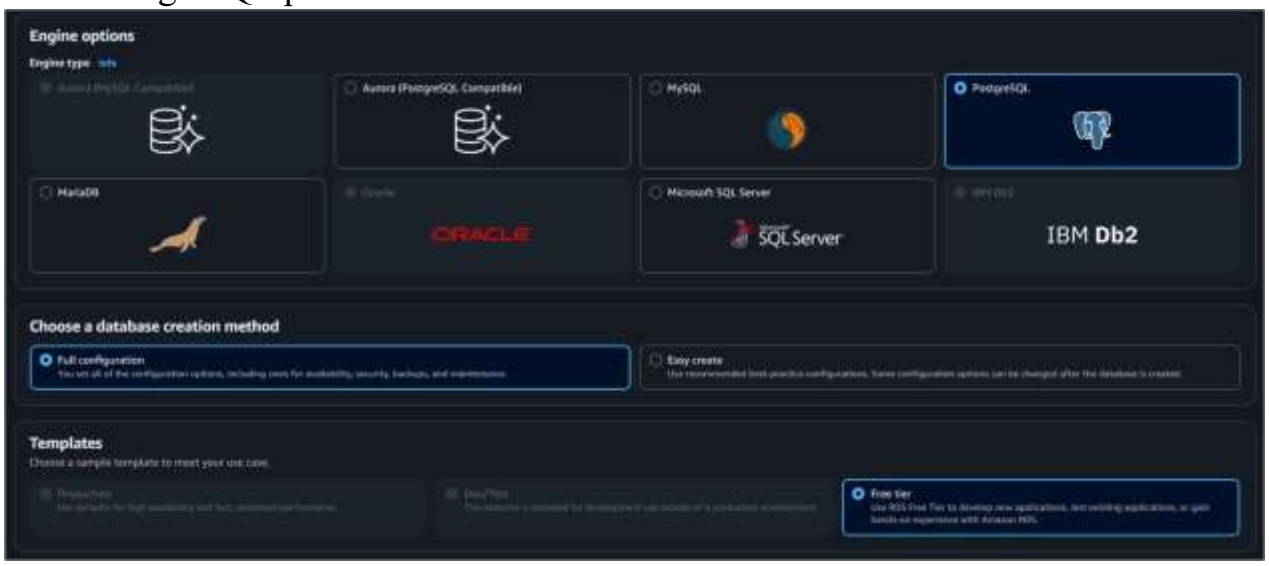
| Feature | Purpose |
|---|---|
| CloudWatch | Performance and health monitoring |
| Automated Backups | Disaster recovery |
| Manual Snapshots | User-created backups / long-term retention |
| Multi-AZ Deployment | High availability using a standby replica in another AZ |
| Read Replicas | Read-only copies for load balancing |
| Security (IAM, KMS, SSL) | Data protection, access control, compliance and safety |
| Performance Insights | SQL/load analysis and performance tuning |
| Storage Auto Scaling | Automatically grows storage as needed |
| Enhanced Monitoring | OS-level real-time metrics |
| Cross-Region Replicas | Replication for global availability |
| Parameter / Option Groups | Database configuration controls |

Steps to create and configure the DB –

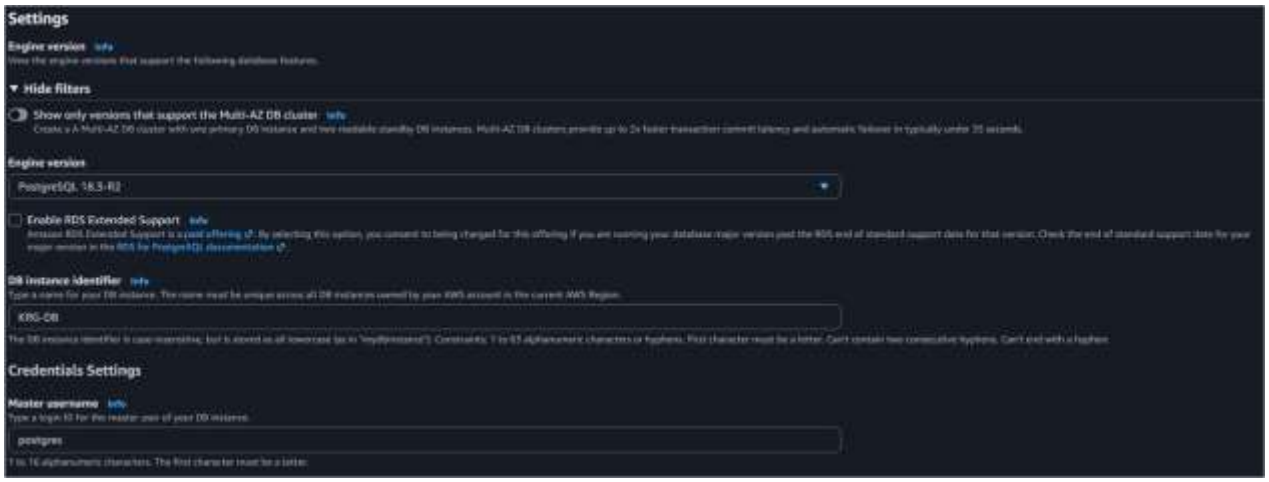
Go to AWS RDS Homepage:



Select PostgreSQL provider:



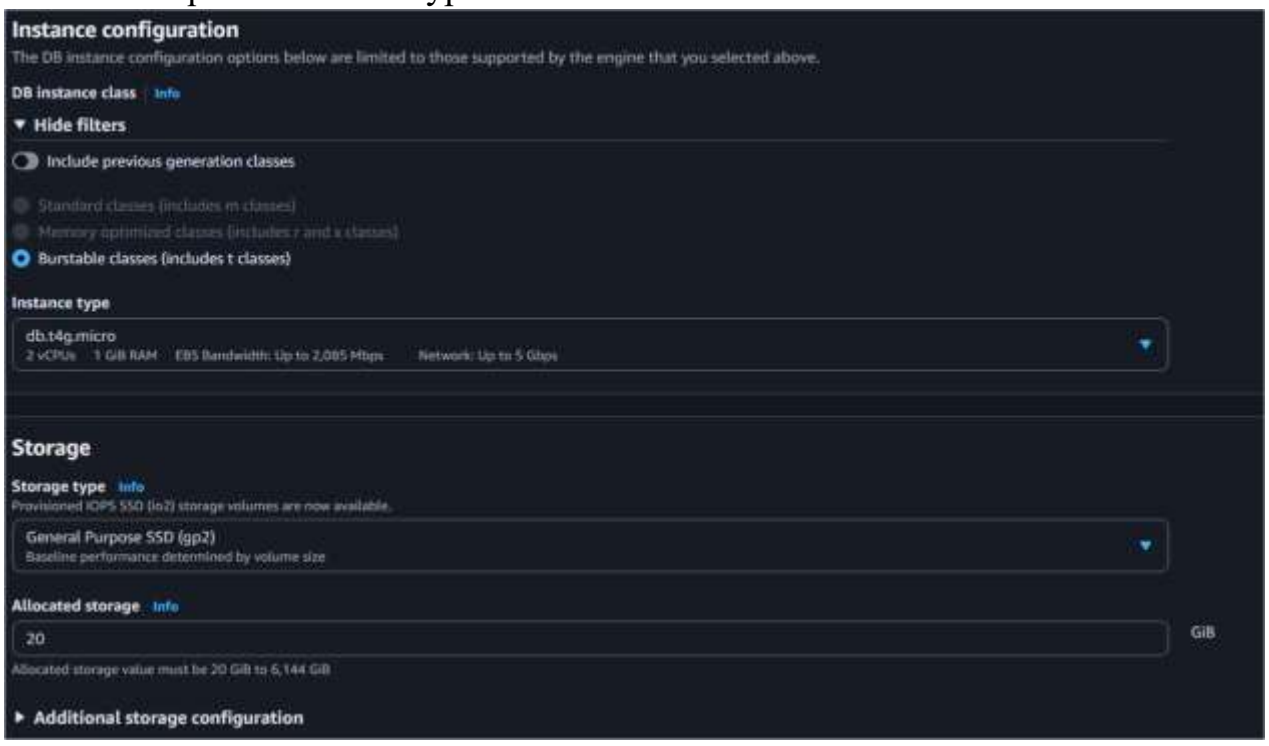

Set DB Instance Identifier:




Select Slef-Managed and set Master Password



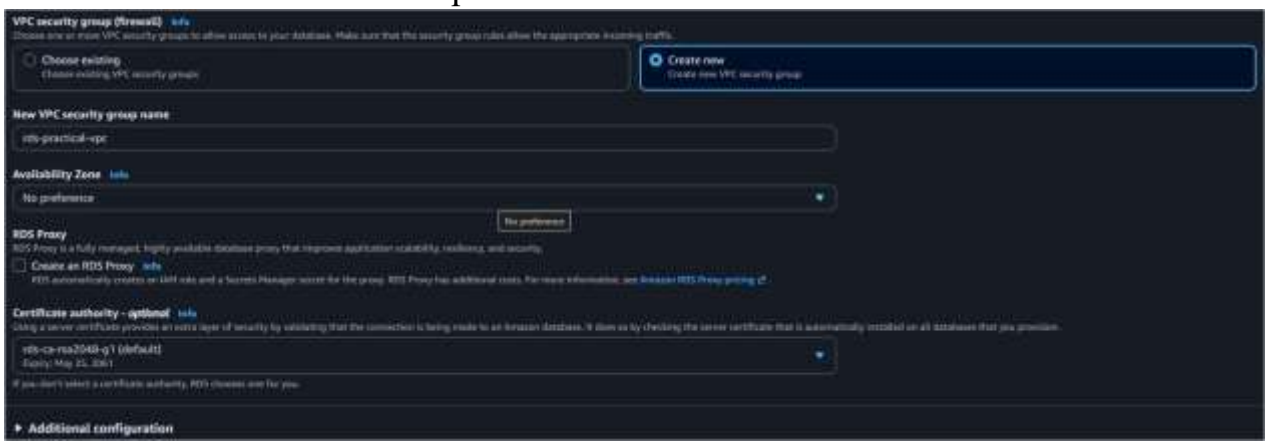

Select the required Instance type:




Allow Public access:



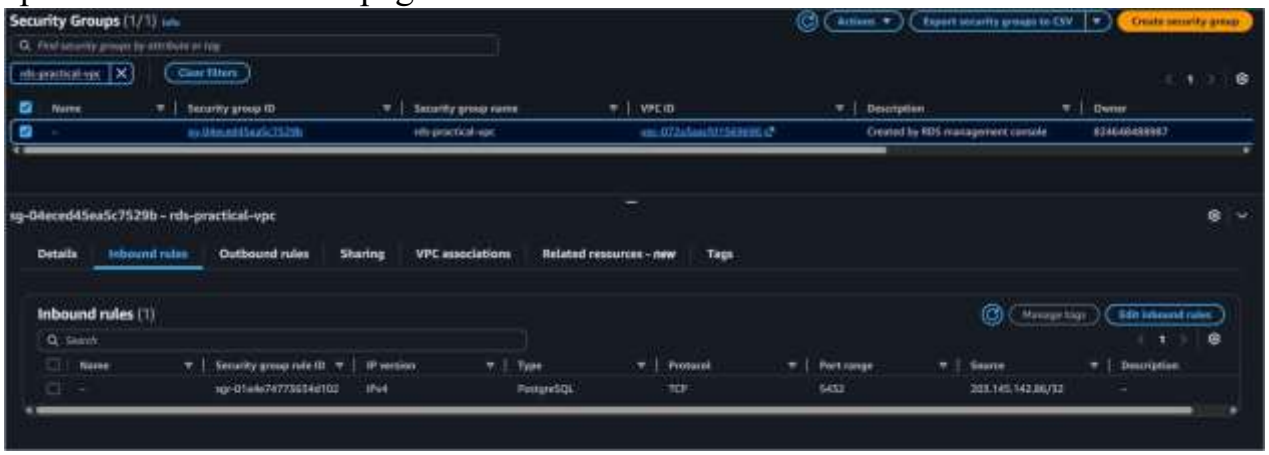
Create custom VPC rules if required:




Wait for the DB to start



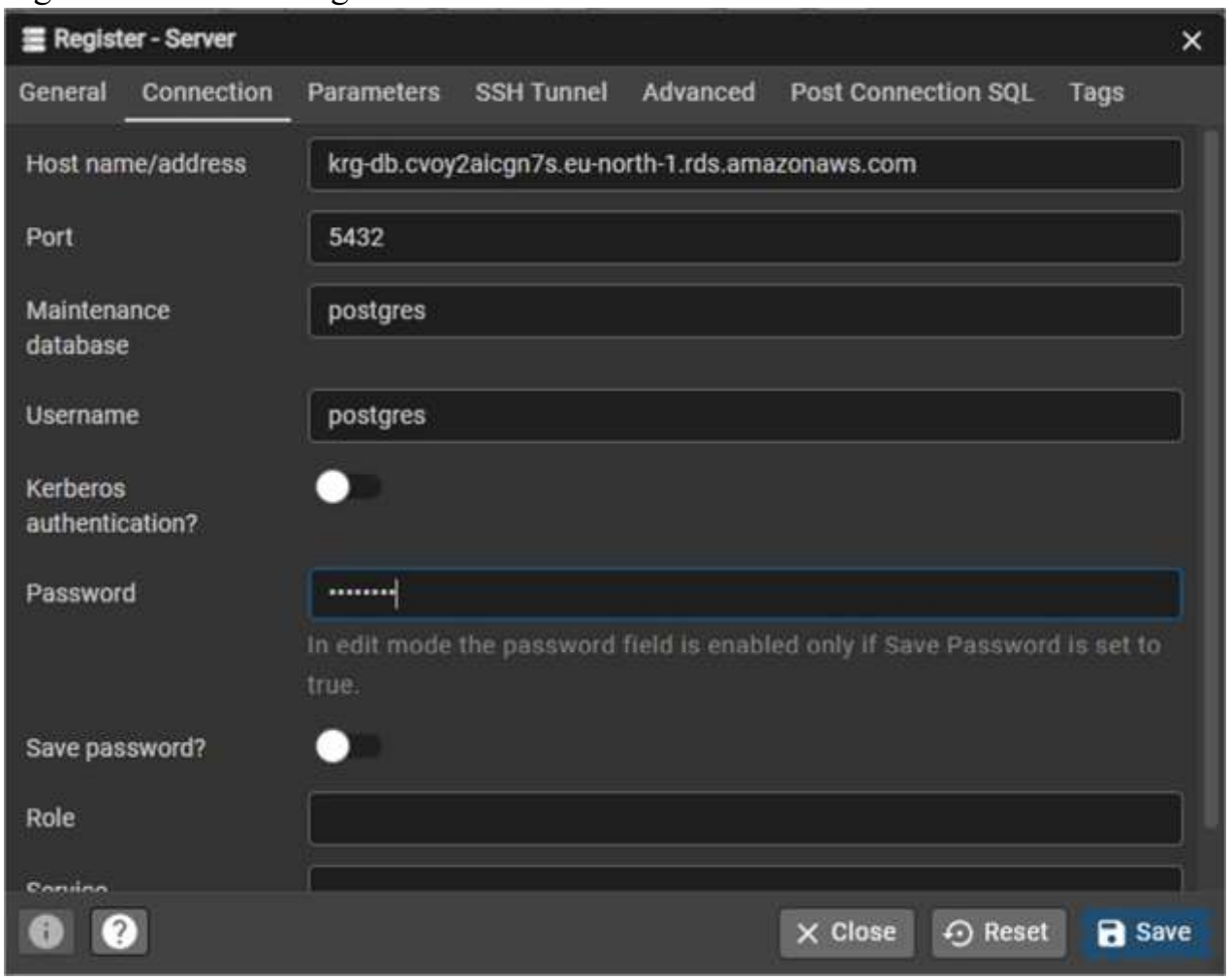

Open the InBound rules page:




Add the host IP address




Register the server in PgAdmin:




The DB is added successfully:




Verify the connection:

